Temperature Control of a Heat Exchanger with Transport Delay

PI versus PI with Smith Predictor, with System Identification in MATLAB

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1. Objective

Processes with a long transport delay are hard to control with a plain PI controller, because the delay forces a conservative tuning. This project identifies a heat-exchanger model from step-test data, designs a PI controller and a PI + Smith predictor, and compares them on setpoint tracking, disturbance rejection, robustness to model mismatch and noise sensitivity.

2. Plant and System Identification

The heat exchanger is modelled as a first-order-plus-dead-time (FOPDT) process, G(s) = K·e^(−θs)/(τs+1), with K = 1.5 °C per unit input, τ = 50 s and θ = 15 s. An RTD sensor adds a 5 s first-order lag, and measurement noise of 0.05 °C is included.

A noisy step test (1 s sampling) was generated and the model was identified with MATLAB System Identification Toolbox (procest, type 'P1D'). The identified parameters match the true values and the model fit is 98.94%.

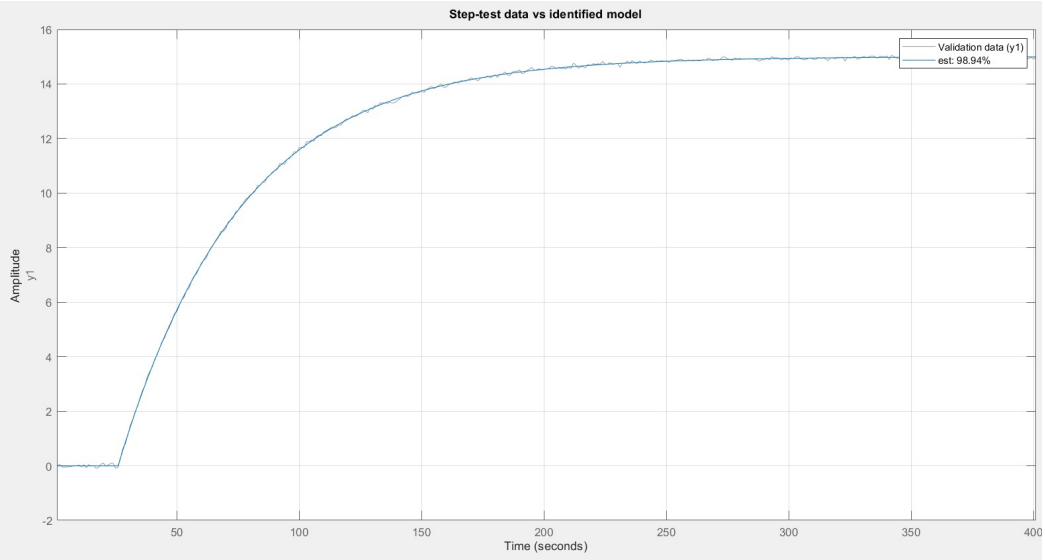


Figure 1. Step-test data versus identified model (fit 98.94%)

Parameter	True	Identified
Gain K	1.5	1.500
Time constant τ (s)	50	50.0
Delay θ (s)	15	15.0

3. Controller Design

3.1 PI controller

Tuned with the SIMC rules. The sensor lag is added to the delay (effective delay θe = θ + 5 = 20 s), which gives Kc = 0.834 and Ti = 50 s.

3.2 PI + Smith predictor

The Smith predictor uses an internal model with and without the delay (the model includes the sensor lag). The controller acts on the error corrected by the difference between the two model outputs, so the delay is removed from the loop's characteristic equation for a perfect model. The PI inside it is tuned more aggressively: Kc = 3.336 and Ti = 50 s.

4. Results

All cases use a 10 °C setpoint step. Overshoot, settling time (5% band, 0.5 °C), setpoint IAE and the peak deviation after a 5% load step at the plant input are reported. Robustness cases change the real plant delay or gain by ±30% while the controller keeps the nominal model.

Case	Controller	Overshoot (%)	Settling (s)	IAE (setpoint)	Dist. peak (°C)
Nominal	PI	4.39	63	387.96	3.45
	PI + Smith	6.78	45.5	221.93	2.70
Delay +30%	PI	12.68	124	468.68	3.73
	PI + Smith	27.83	102.5	342.33	3.07
Delay −30%	PI	0.09	72	350.69	3.16
	PI + Smith	0.64	50.5	201.11	2.29
Gain +30%	PI	15.73	99.5	376.67	4.22
	PI + Smith	36.21	74.5	279.56	3.36
Gain −30%	PI	0	112.5	521.41	2.63
	PI + Smith	0	69.5	307.06	2.03

Table 1. Final design (sensor lag modelled in both controllers).

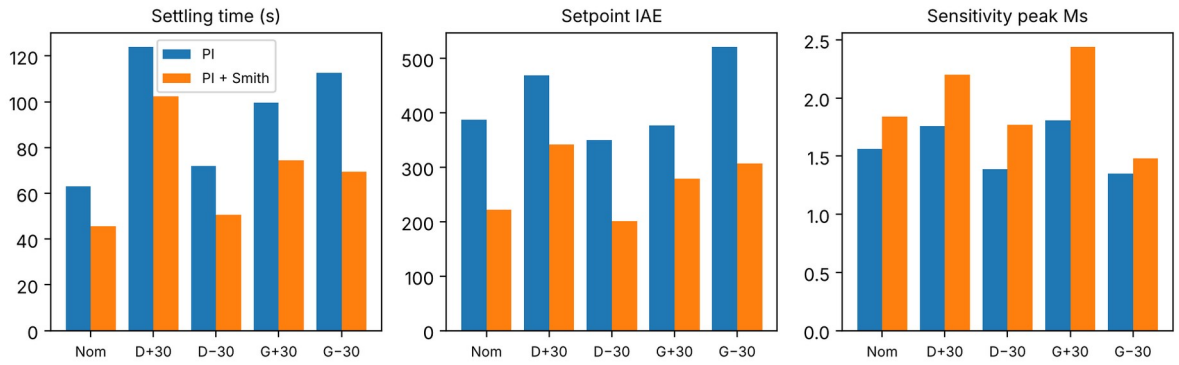


Figure 2. Settling time, setpoint IAE and sensitivity peak Ms for each case

4.1 Findings

- Settling time is 17–38% shorter with the Smith predictor in every case.
- Setpoint IAE is about 26–43% lower with the Smith predictor.
- Disturbance peak is about 18–28% lower with the Smith predictor.
- Overshoot is higher with the Smith predictor in every case that overshoots, and more than double for Delay +30% (12.7% to 27.8%) and Gain +30% (15.7% to 36.2%).

4.2 Robustness and noise

Case	Ms (PI)	Ms (PI + Smith)
Nominal	1.56	1.84
Delay +30%	1.76	2.20

Delay −30%	1.39	1.77
Gain +30%	1.81	2.44
Gain −30%	1.35	1.48

Table 2. Sensitivity peak Ms (rule of thumb: 1.2–2).

The Smith predictor has a higher Ms in every case, and exceeds 2 for Delay +30% (2.20) and Gain +30% (2.44), so its margin is thinner when the model is wrong. Noise on the measurement produces valve movement (standard deviation) of 0.040 for PI and 0.156 for PI + Smith, about 3.9 times more, because of its higher controller gain.

5. Discussion and Limitations

- The two controllers are not tuned for equal robustness. The Smith design is more aggressive (higher Ms), so part of its speed advantage comes from tuning, not only from the predictor structure. A retune of both at the same Ms would give a fairer comparison.
- The Smith predictor's benefit depends on model accuracy. With a +30% delay or gain error its overshoot grows sharply.
- Adding the sensor lag to the design changed the results substantially compared with the first design (Appendix A), which shows how much unmodelled dynamics matter.
- The study is simulation-only, with a known simulated plant. A real heat exchanger would add nonlinearity and valve limits (not modelled).
- Disturbance IAE is not used as a metric: with integral action it is nearly the same for any controller with the same Ti and gain ratio, so it carries little information.

6. Conclusion

The Smith predictor improved settling time, setpoint IAE and disturbance rejection for this delay-dominant process, at the cost of higher overshoot under model mismatch, a higher sensitivity peak and about four times more valve activity from noise. It is worth using when the delay is well known and the actuator can tolerate the noise; PI alone is the safer choice when the model is uncertain.

Appendix A. First design (sensor lag not in the controller model)

Case	Controller	Overshoot (%)	Settling (s)	IAE (setpoint)	Dist. peak (°C)
Nominal	PI	17.15	98.5	378.47	3.23
	PI + Smith	13.98	74.5	282.09	2.81
Delay +30%	PI	30.88	163	519.84	3.54
	PI + Smith	31.80	130.5	422.93	3.16
Delay −30%	PI	5.62	76	288.01	2.91
	PI + Smith	1.12	38.5	205.51	2.43
Gain +30%	PI	34.97	132	432.28	3.97
	PI + Smith	34.01	99.5	320.45	3.46
Gain −30%	PI	2.49	69.5	401	2.45
	PI + Smith	0.81	58	312.99	2.15

Table A1. First design. Nominal Ms was 1.84 (PI) versus 1.81 (Smith); valve noise 0.056 versus 0.151.

Appendix B. Tools

MATLAB with Control System Toolbox and System Identification Toolbox. Scripts: hx_model, hx_identify, hx_pi, hx_smith, hx_robust, hx_dist, hx_noise, hx_final, hx_final2, hx_check.